Ordinary Differential Equations: Abridged Notes

Chris Ritter

September 10, 2026

These abridged notes were reconstructed afterward because I did not take direct notes during class. They are not a record of the actual lecture.

Contents

1 Topic Summary **2**

2 Phase Lines and Stability **2**

2.1 Example: Quadratic Population Rate 2

2.2 Example: Three Equilibria 2

1 Topic Summary

An autonomous equation has the form $y' = f(y)$: its rate depends on the current state, not explicitly on time. Its equilibria are the roots of f. A phase line records the sign of f between consecutive equilibria and shows whether nearby solutions move toward or away from each one. This gives qualitative information about population models without solving the equation.

2 Phase Lines and Stability

To analyze $y' = f(y)$, first solve $f(y) = 0$. On each interval between equilibrium values, test one point to find the sign of f: positive means solutions increase; negative means they decrease. An equilibrium is **asymptotically stable** if nearby solutions approach it as time increases, and **unstable** if nearby solutions move away. When $f'(y_*) \neq 0$, the sign of $f'(y_*)$ gives the same local test: negative is stable and positive is unstable.

2.1 Example: Quadratic Population Rate

Adapted from Boyce et al., 11th ed., §2.5, Problem 1. Consider $y' = ay + by^2$ with $a, b > 0$.

Solution.

Factor $f(y) = y(a + by)$. The equilibria are $y = -a/b$ and $y = 0$. The sign chart is

y	$(-\infty, -a/b)$	$(-a/b, 0)$	$(0, \infty)$
$f(y)$	$+$	$-$	$+$

Thus arrows point toward $-a/b$ from both sides, while they point away from 0. Therefore

$$\boxed{-a/b \text{ is asymptotically stable, and 0 is unstable.}}$$

For nonnegative population, $y = 0$ is the only equilibrium in the physical range; every positive solution initially increases. The negative equilibrium is mathematically valid but not a population.

2.2 Example: Three Equilibria

Adapted from Boyce et al., 11th ed., §2.5, Problem 2. Consider $y' = y(y - 1)(y - 2)$ for $y \geq 0$.

Solution.

The equilibria are 0, 1, and 2. Testing one value in each interval gives

y	$(-\infty, 0)$	$(0, 1)$	$(1, 2)$	$(2, \infty)$
$f(y)$	$-$	$+$	$-$	$+$

The arrows on $(0, 1)$ point right and those on $(1, 2)$ point left, so nearby solutions approach 1. On either side of 0 and 2, nearby solutions move away. Hence

$$\boxed{1 \text{ is asymptotically stable; 0 and 2 are unstable.}}$$

For physical initial data $y_0 \geq 0$, the negative part of the phase line is only a mathematical extension.